

1. What is the mass of 4 moles of carbon dioxide?

Given: $m_c = 12$ g, $m_o = 16$ g, **Find:** m .

$$\begin{aligned} m &= 4 \cdot (m_c + (2 \cdot m_o)) \\ &= 4 \cdot (12 \text{ g} + (2 \cdot 16 \text{ g})) \\ &= 4 \cdot 44 \text{ g}. \end{aligned}$$

$$\boxed{= 176 \text{ g}}$$

2. A typical bacterium has a mass of 1 pg. Calculate the mass of a mole of bacteria.

Given: $m_b = 1$ pg, $m_e = 5000$ kg, **Find:** m .

$$\begin{aligned} m &= N_A \cdot m_b \\ &= 6.022 \cdot 10^{23} \cdot 1 \text{ pg} \\ &= 6.022 \cdot 10^8 \text{ kg}. \end{aligned}$$

$$\boxed{\approx 120000 \text{ elephant-masses}}$$

3. Convert the following temperatures into Kelvin:

(a) Boiling point of hydrogen: -252.87 degrees Celsius.

$$-252.87 \text{ }^\circ\text{C} = 273.15 + (-252.87) \text{ K}.$$

$$\boxed{= 20.28 \text{ K}}$$

(b) Triple point of water: 0.01 degrees Celsius.

$$0.01 \text{ }^\circ\text{C} = 273.15 + (0.01) \text{ K}.$$

$$\boxed{= 273.16 \text{ K}}$$

4. Convert the following temperatures into Fahrenheit:

(a) -13 degrees Celsius.

$$-13 \text{ }^\circ\text{C} = -13 \cdot 1.8 + 32 \text{ }^\circ\text{F}.$$

$$\boxed{= 8.6 \text{ }^\circ\text{F}}$$

(b) 25 degrees Kelvin.

$$\begin{aligned} 25 \text{ K} &= 25 - 273.15 \text{ }^\circ\text{C} \\ &= -248.15 \text{ }^\circ\text{C} \\ -248.15 \text{ }^\circ\text{C} &= -248.15 \cdot 1.8 + 32 \text{ }^\circ\text{F} \\ &= -414.67 \text{ }^\circ\text{F} \end{aligned}$$

$$\boxed{\approx -410 \text{ }^\circ\text{F}}$$

5. A system contains n atoms, each of which can only have zero or one quanta of energy. How many ways can you arrange r quanta of energy when:

(a) $n = 2, r = 1$

$$\binom{2}{1} = \boxed{2}.$$

(b) $n = 20, r = 10$

$$\binom{20}{10} = \boxed{184756}.$$

(c) $n = 2 \cdot 10^{23}, r = 10^{23}$

$$\boxed{\binom{2 \cdot 10^{23}}{10^{23}}}.$$